

# Elementary Number Theory Exercises

Q&A Notes

## Question 1

Prove that if  $a > b > 0$  and  $c = a + b$ , then  $c \bmod a = b$ .

## Answer 1

Since  $c = a + b$ , write  $c$  in division-theorem form with divisor  $a$ :

$$c = 1 \cdot a + b.$$

Because  $a > b > 0$ , we have

$$0 \leq b < a.$$

So  $b$  is the unique remainder when  $c$  is divided by  $a$ . Therefore,

$$\boxed{c \bmod a = b.}$$

## Question 2

Prove that there are infinitely many primes.

## Answer 2

Assume, for contradiction, that there are only finitely many primes:

$$p_1, p_2, \dots, p_k.$$

Form the integer

$$N = p_1 p_2 \cdots p_k + 1.$$

For each  $i$ ,

$$N \bmod p_i = 1,$$

so no  $p_i$  divides  $N$ . But  $N > 1$ , so  $N$  must have some prime divisor. That prime divisor is not in the list, contradicting the assumption that the list contained all primes.

Therefore,

$$\boxed{\text{there are infinitely many primes.}}$$

## Question 3

Prove that if  $a \mid b$  and  $b \mid c$ , then  $a \mid c$ .

## Answer 3

Since  $a \mid b$ , there is an integer  $m$  such that

$$b = am.$$

Since  $b \mid c$ , there is an integer  $n$  such that

$$c = bn.$$

Substitute  $b = am$ :

$$c = (am)n = a(mn).$$

Because  $mn \in \mathbb{Z}$ , this means

$$a \mid c.$$

#### Question 4

Prove that if  $p$  is prime and  $0 < k < p$ , then  $\gcd(k, p) = 1$ .

#### Answer 4

Because  $p$  is prime, its only positive divisors are

$$1 \quad \text{and} \quad p.$$

Any common divisor of  $k$  and  $p$  must divide  $p$ , so it must be either 1 or  $p$ .

But  $p \nmid k$ , since  $0 < k < p$ . Therefore the only possible positive common divisor is 1. Hence

$$\gcd(k, p) = 1.$$

#### Question 5

Prove Corollary 31.5: for all positive integers  $n, a, b$ , if  $n \mid ab$  and  $\gcd(a, n) = 1$ , then  $n \mid b$ .

#### Answer 5

Since  $\gcd(a, n) = 1$ , Bézout's theorem gives integers  $x, y$  such that

$$ax + ny = 1.$$

Multiply both sides by  $b$ :

$$abx + nby = b.$$

Now  $n \mid ab$ , so  $n \mid abx$ . Also,

$$nby = n(by),$$

so  $n \mid nby$ . Therefore  $n$  divides the sum:

$$n \mid (abx + nby).$$

Since  $abx + nby = b$ , we get

$$n \mid b.$$

#### Question 6

Prove that if  $p$  is prime and  $0 < k < p$ , then  $p \mid \binom{p}{k}$ . Conclude that for all integers  $a, b$  and all primes  $p$ ,

$$(a + b)^p \equiv a^p + b^p \pmod{p}.$$

#### Answer 6

First,

$$\binom{p}{k} = \frac{p!}{k!(p-k)!}.$$

Equivalently,

$$k!(p-k)! \binom{p}{k} = p!.$$

The right-hand side contains a factor  $p$ , so

$$p \mid k!(p-k)! \binom{p}{k}.$$

For  $0 < k < p$ , none of the integers in  $k!$  or  $(p-k)!$  is divisible by  $p$ . Since  $p$  is prime, Euclid's lemma forces

$$p \mid \binom{p}{k}.$$

Now use the binomial theorem:

$$(a+b)^p = \sum_{k=0}^p \binom{p}{k} a^{p-k} b^k.$$

The first and last terms are

$$a^p \quad \text{and} \quad b^p.$$

Every middle coefficient  $\binom{p}{k}$ , where  $1 \leq k \leq p-1$ , is divisible by  $p$ . Hence all middle terms vanish modulo  $p$ . Therefore,

$$(a+b)^p \equiv a^p + b^p \pmod{p}.$$

### Question 7

Prove that if  $a$  and  $b$  are positive integers such that  $a \mid b$ , then

$$(x \bmod b) \bmod a = x \bmod a$$

for any  $x$ . Under the same assumptions, prove that

$$x \equiv y \pmod{b} \implies x \equiv y \pmod{a}.$$

### Answer 7

Let

$$x = qb + r, \quad 0 \leq r < b.$$

Then

$$x \bmod b = r.$$

Since  $a \mid b$ , write

$$b = at$$

for some integer  $t$ . Then

$$x = qat + r.$$

So  $x$  and  $r$  differ by a multiple of  $a$ :

$$x - r = qat.$$

Therefore,

$$x \equiv r \pmod{a}.$$

Thus  $x$  and  $x \bmod b$  have the same remainder modulo  $a$ , so

$$(x \bmod b) \bmod a = x \bmod a.$$

For the second part, if

$$x \equiv y \pmod{b},$$

then

$$b \mid (x - y).$$

Since  $a \mid b$ , transitivity of divisibility gives

$$a \mid (x - y).$$

Therefore,

$$x \equiv y \pmod{a}.$$

### Question 8

For any integer  $k > 0$ , an integer  $n$  is a  $k$ th power if there exists an integer  $a$  such that  $a^k = n$ . Furthermore,  $n > 1$  is a nontrivial power if it is a  $k$ th power for some integer  $k > 1$ . Show how to determine whether a given  $\beta$ -bit integer  $n$  is a nontrivial power in time polynomial in  $\beta$ .

### Answer 8

It is enough to test exponents

$$2 \leq k \leq \beta,$$

because if  $n = a^k$  and  $a \geq 2$ , then

$$2^k \leq a^k = n < 2^\beta,$$

so  $k < \beta$ .

For each  $k = 2, 3, \dots, \beta$ , use binary search to find whether there is an integer  $a$  such that

$$a^k = n.$$

The possible range for  $a$  is

$$2 \leq a \leq 2^{\lceil \beta/k \rceil}.$$

During binary search, compute  $a^k$  using repeated squaring and compare it with  $n$ .

There are at most  $\beta$  exponents to test, each binary search has polynomially many bit operations, and exponentiation/comparison on  $\beta$ -bit integers is polynomial in  $\beta$ . Therefore the whole procedure runs in polynomial time.

Test all  $2 \leq k \leq \beta$  using integer  $k$ th-root search.

### Question 9

Prove the following basic gcd identities:

$$\gcd(a, b) = \gcd(b, a),$$

$$\gcd(a, b) = \gcd(-a, b),$$

$$\gcd(a, b) = \gcd(|a|, |b|),$$

$$\gcd(a, 0) = |a|,$$

$$\gcd(a, ka) = |a| \quad \text{for any } k \in \mathbb{Z}.$$

### Answer 9

The first identity holds because common divisors are symmetric:

$$d \mid a \text{ and } d \mid b \iff d \mid b \text{ and } d \mid a.$$

Thus

$$\boxed{\gcd(a, b) = \gcd(b, a).}$$

The second holds because

$$d \mid a \iff d \mid (-a).$$

So  $a$  and  $-a$  have the same divisors:

$$\boxed{\gcd(a, b) = \gcd(-a, b).}$$

The third follows by applying the sign rule to both arguments:

$$\boxed{\gcd(a, b) = \gcd(|a|, |b|).}$$

For the fourth, every positive divisor of  $a$  divides 0, since

$$0 = 0 \cdot d.$$

The largest positive divisor of  $a$  is  $|a|$ , so

$$\boxed{\gcd(a, 0) = |a|}.$$

For the fifth,  $|a| \mid a$  and  $|a| \mid ka$ , so  $|a|$  is a common divisor. No common divisor of  $a$  and  $ka$  can be larger than  $|a|$ , because it must divide  $a$ . Therefore,

$$\boxed{\gcd(a, ka) = |a|}.$$

### Question 10

Show that the gcd operator is associative. That is, prove that for all integers  $a, b, c$ ,

$$\gcd(a, \gcd(b, c)) = \gcd(\gcd(a, b), c).$$

### Answer 10

Let

$$g_1 = \gcd(a, \gcd(b, c)).$$

Then  $g_1 \mid a$  and  $g_1 \mid \gcd(b, c)$ . Since  $g_1 \mid \gcd(b, c)$ , we have

$$g_1 \mid b \quad \text{and} \quad g_1 \mid c.$$

So  $g_1$  is a common divisor of  $a, b, c$ .

Conversely, any common divisor of  $a, b, c$  divides  $b$  and  $c$ , so it divides  $\gcd(b, c)$ . If it also divides  $a$ , then it divides  $\gcd(a, \gcd(b, c))$ . Therefore  $g_1$  is exactly the greatest common divisor of all three numbers.

Similarly,

$$g_2 = \gcd(\gcd(a, b), c)$$

is also the greatest common divisor of  $a, b, c$ . Hence

$$\gcd(a, \gcd(b, c)) = \gcd(\gcd(a, b), c).$$

### Question 11

Prove Theorem 31.8: unique factorization.

### Answer 11

Assume  $n > 1$  has two prime factorizations:

$$n = p_1 p_2 \cdots p_r = q_1 q_2 \cdots q_s.$$

Since  $p_1 \mid n$ , we have

$$p_1 \mid q_1 q_2 \cdots q_s.$$

By Euclid's lemma,  $p_1$  divides some  $q_j$ . Since both are prime, this forces

$$p_1 = q_j.$$

Move  $q_j$  to the front of the right-hand product and cancel it with  $p_1$ . Then repeat the same argument for  $p_2, p_3, \dots$

Every prime factor on the left must match a prime factor on the right. Neither side can have extra unmatched primes, because then a product of primes greater than 1 would have to equal 1.

Thus the two factorizations contain exactly the same primes, counted with repetition. Grouping repeated primes gives the same prime powers:

$$n = p_1^{e_1} p_2^{e_2} \cdots p_r^{e_r} \text{ uniquely.}$$

### Question 12

Give efficient algorithms for dividing a  $\beta$ -bit integer by a shorter integer and for taking the remainder of a  $\beta$ -bit integer when divided by a shorter integer. The algorithms should run in time  $\Theta(\beta^2)$ .

### Answer 12

Use binary long division.

Let  $N$  be the  $\beta$ -bit dividend and let  $d$  be the shorter positive divisor. Scan the bits of  $N$  from most significant to least significant. Maintain a partial remainder  $r$ .

```
q = 0
r = 0
for each bit bit_i of N from left to right:
    r = 2*r + bit_i
    q = 2*q
    if r >= d:
        r = r - d
        q = q + 1
return q, r
```

At each step, the algorithm shifts  $r$ , compares with  $d$ , and possibly subtracts  $d$ . These opera-

tions cost  $O(\beta)$  bit operations. There are  $\beta$  bits, so the total running time is

$$\boxed{\Theta(\beta^2)}.$$

The quotient is  $q$ , and the remainder is  $r = N \bmod d$ .

### Question 13

Give an efficient algorithm to convert a given  $\beta$ -bit binary integer to a decimal representation. Argue that if multiplication or division of integers whose length is at most  $\beta$  takes time  $M(\beta)$ , then binary-to-decimal conversion takes time  $\Theta(M(\beta) \lg \beta)$ .

### Answer 13

Use divide and conquer.

Split the binary integer  $N$  into two halves:

$$N = H \cdot 2^m + L,$$

where  $L$  contains the lower  $m$  bits and  $H$  contains the remaining upper bits.

Recursively convert  $H$  and  $L$  to decimal. Then combine:

$$N = H \cdot 2^m + L.$$

The multiplication by  $2^m$  and the addition are performed on integers of length at most  $\beta$ , costing  $O(M(\beta))$  at the top level.

The recurrence is

$$T(\beta) = 2T(\beta/2) + O(M(\beta)).$$

Across each recursion level, the total multiplication/addition cost is  $O(M(\beta))$ , and there are  $O(\lg \beta)$  levels. Therefore,

$$\boxed{T(\beta) = \Theta(M(\beta) \lg \beta)}.$$

# Euclidean Algorithm Exercises

Q&A Notes

## Question 1

Suppose

$$a = p_1^{e_1} p_2^{e_2} \cdots p_r^{e_r} \quad \text{and} \quad b = p_1^{f_1} p_2^{f_2} \cdots p_r^{f_r},$$

where missing primes are given exponent 0. Prove that

$$\gcd(a, b) = p_1^{\min(e_1, f_1)} p_2^{\min(e_2, f_2)} \cdots p_r^{\min(e_r, f_r)}.$$

## Answer 1

The gcd takes exactly the smaller exponent of each prime.

Let  $d$  be any common divisor of  $a$  and  $b$ . If

$$d = p_1^{g_1} p_2^{g_2} \cdots p_r^{g_r},$$

then  $d \mid a$  gives  $g_i \leq e_i$ , and  $d \mid b$  gives  $g_i \leq f_i$ . Hence

$$g_i \leq \min(e_i, f_i)$$

for every  $i$ . Therefore every common divisor divides

$$p_1^{\min(e_1, f_1)} p_2^{\min(e_2, f_2)} \cdots p_r^{\min(e_r, f_r)}.$$

On the other hand, this number divides both  $a$  and  $b$ , because each exponent is no larger than the corresponding exponent in  $a$  and in  $b$ . So it is the greatest common divisor:

$$\gcd(a, b) = \prod_{i=1}^r p_i^{\min(e_i, f_i)}.$$

## Question 2

Compute the values  $(d, x, y)$  returned by

ExtendedEuclid(899, 493).

## Answer 2

The result is

$$(d, x, y) = (29, -6, 11).$$

So

$$29 = 899(-6) + 493(11).$$

Use the Euclidean algorithm:

$$899 = 493(1) + 406,$$

$$493 = 406(1) + 87,$$

$$406 = 87(4) + 58,$$

$$87 = 58(1) + 29,$$

$$58 = 29(2) + 0.$$

Thus

$$\gcd(899, 493) = 29.$$

Now back substitute:

$$29 = 87 - 58,$$

$$58 = 406 - 4(87),$$

so

$$29 = 87 - (406 - 4(87)) = 5(87) - 406.$$

Also

$$87 = 493 - 406,$$

so

$$29 = 5(493 - 406) - 406 = 5(493) - 6(406).$$

Finally,

$$406 = 899 - 493,$$

hence

$$29 = 5(493) - 6(899 - 493) = -6(899) + 11(493).$$

Therefore `ExtendedEuclid(899, 493)` returns

$$(29, -6, 11).$$

### Question 3

Prove that for all integers  $a, k, n$ ,

$$\gcd(a, n) = \gcd(a + kn, n).$$

### Answer 3

The two gcds are equal because  $a$  and  $a + kn$  have exactly the same common divisors with  $n$ . Let  $d \mid a$  and  $d \mid n$ . Since  $d \mid n$ , we have  $d \mid kn$ . Therefore

$$d \mid a + kn.$$

So every common divisor of  $a$  and  $n$  is also a common divisor of  $a + kn$  and  $n$ .

Conversely, let  $d \mid a + kn$  and  $d \mid n$ . Again  $d \mid kn$ , so

$$d \mid (a + kn) - kn = a.$$

Thus every common divisor of  $a + kn$  and  $n$  is also a common divisor of  $a$  and  $n$ .

The two pairs have the same common divisors, so their greatest common divisors are equal:

$$\gcd(a, n) = \gcd(a + kn, n).$$

### Question 4

Rewrite Euclid's algorithm in an iterative form that uses only a constant amount of memory.

### Answer 4

Keep only the current pair  $(a, b)$  and one temporary remainder.

### Iterative Euclid

```
EuclidIterative(a, b):  
    while b != 0:  
        r = a mod b  
        a = b  
        b = r  
    return a
```

At each loop iteration, the pair

$$(a, b)$$

is replaced by

$$(b, a \bmod b).$$

The gcd does not change because

$$\gcd(a, b) = \gcd(b, a \bmod b).$$

Only  $a$ ,  $b$ , and  $r$  are stored, so the memory usage is constant:

$$O(1) \text{ integer variables.}$$

### Question 5

If  $a > b \geq 0$ , show that  $\text{Euclid}(a, b)$  makes at most

$$1 + \log_{\phi} b$$

recursive calls. Improve this bound to

$$1 + \log_{\phi} \left( \frac{b}{\gcd(a, b)} \right).$$

### Answer 5

Let  $\phi = (1 + \sqrt{5})/2$ . The key fact is Lamé's Fibonacci bound:  
If  $a > b \geq 1$  and  $\text{Euclid}(a, b)$  makes  $k \geq 1$  recursive calls, then

$$b \geq F_{k+1}.$$

Also, Fibonacci numbers grow exponentially:

$$F_{k+1} \geq \phi^{k-1}.$$

Therefore,

$$b \geq F_{k+1} \geq \phi^{k-1}.$$

Taking  $\log_{\phi}$  gives

$$\log_{\phi} b \geq k - 1,$$

so

$$k \leq 1 + \log_{\phi} b.$$

For the improved bound, let

$$g = \gcd(a, b).$$

Dividing both inputs by  $g$  does not change the sequence of quotients in Euclid's algorithm:

$$\text{Euclid}(a, b) \quad \text{and} \quad \text{Euclid}\left(\frac{a}{g}, \frac{b}{g}\right)$$

make the same number of recursive calls. Applying the previous bound to the reduced pair gives

$$k \leq 1 + \log_\phi\left(\frac{b}{g}\right) = 1 + \log_\phi\left(\frac{b}{\gcd(a, b)}\right).$$

### Question 6

What does

$$\text{ExtendedEuclid}(F_{k+1}, F_k)$$

return? Prove your answer correct.

### Answer 6

For  $k \geq 2$ , the return value is

$$\left(1, (-1)^{k+1}F_{k-2}, (-1)^kF_{k-1}\right).$$

That means

$$F_{k+1}((-1)^{k+1}F_{k-2}) + F_k((-1)^kF_{k-1}) = 1.$$

The gcd part is 1, since consecutive Fibonacci numbers are relatively prime:

$$\gcd(F_{k+1}, F_k) = 1.$$

For the coefficients, use this form of Cassini's identity:

$$F_k F_{k-1} - F_{k+1} F_{k-2} = (-1)^k.$$

Multiply both sides by  $(-1)^k$ :

$$(-1)^{k+1}F_{k+1}F_{k-2} + (-1)^kF_kF_{k-1} = 1,$$

which gives the coefficients

$$x = (-1)^{k+1}F_{k-2}, \quad y = (-1)^kF_{k-1}.$$

For example, when  $k = 5$ ,

$$\text{ExtendedEuclid}(F_6, F_5) = \text{ExtendedEuclid}(8, 5)$$

returns

$$(1, 2, -3),$$

because

$$8(2) + 5(-3) = 1.$$

### Question 7

Define the gcd function for more than two arguments by

$$\gcd(a_0, a_1, \dots, a_n) = \gcd(a_0, \gcd(a_1, \dots, a_n)).$$

Show that the result is independent of the order of the arguments. Also show how to find integers  $x_0, x_1, \dots, x_n$  such that

$$\gcd(a_0, a_1, \dots, a_n) = a_0x_0 + a_1x_1 + \dots + a_nx_n.$$

### Answer 7

The result is independent of order because the gcd is the largest integer that divides every input.

Let

$$g = \gcd(a_0, a_1, \dots, a_n).$$

Then  $g$  divides every  $a_i$ , and any integer that divides every  $a_i$  must divide  $g$ . This description does not depend on the order in which the inputs are written. Therefore the multi-argument gcd is order independent.

To find coefficients, combine the numbers one at a time using Extended Euclid.

Start with

$$D_0 = a_0.$$

Assume after processing  $a_0, \dots, a_{i-1}$ , we have

$$D_{i-1} = a_0x_0 + a_1x_1 + \dots + a_{i-1}x_{i-1}.$$

Run Extended Euclid on  $D_{i-1}$  and  $a_i$ :

$$D_i = \gcd(D_{i-1}, a_i) = uD_{i-1} + va_i.$$

Substitute the old expression for  $D_{i-1}$ :

$$D_i = u(a_0x_0 + \dots + a_{i-1}x_{i-1}) + va_i.$$

So the old coefficients are multiplied by  $u$ , and the new coefficient of  $a_i$  is  $v$ .

After all inputs are processed,

$$\boxed{\gcd(a_0, \dots, a_n) = a_0x_0 + \dots + a_nx_n.}$$

For the division count, each new argument costs a constant amount unless the current gcd strictly decreases. Whenever the current positive gcd strictly decreases, it becomes a proper divisor and is at most half as large. Hence there can be only  $O(\lg M)$  such decreases, where

$$M = \max\{a_0, a_1, \dots, a_n\}.$$

Thus the total number of divisions is

$$\boxed{O(n + \lg M).}$$

### Question 8

Show how to compute

$$\text{lcm}(a_1, a_2, \dots, a_n)$$

efficiently using the two-argument gcd operation as a subroutine.

### Answer 8

For two positive integers,

$$\text{lcm}(a, b) = \frac{ab}{\text{gcd}(a, b)}.$$

It is usually better to compute it as

$$\text{lcm}(a, b) = \left( \frac{a}{\text{gcd}(a, b)} \right) b$$

to keep the intermediate value smaller.

For many inputs, fold from left to right:

#### LCM using GCD

```

LCM(a1, a2, ..., an):
  L = a1
  for i = 2 to n:
    g = gcd(L, ai)
    L = (L / g) * ai
  return L

```

The invariant is:

$$L = \text{lcm}(a_1, \dots, a_i)$$

after processing  $a_i$ . Therefore, after the loop finishes,

$$L = \text{lcm}(a_1, \dots, a_n).$$

### Question 9

Prove that  $n_1, n_2, n_3, n_4$  are pairwise relatively prime if and only if

$$\text{gcd}(n_1 n_2, n_3 n_4) = 1$$

and

$$\text{gcd}(n_1 n_3, n_2 n_4) = 1.$$

Then explain the general idea for  $n_1, n_2, \dots, n_k$ .

### Answer 9

First suppose  $n_1, n_2, n_3, n_4$  are pairwise relatively prime. Then no prime divides two different  $n_i$ 's. Therefore no prime can divide both products  $n_1 n_2$  and  $n_3 n_4$ , and no prime can divide both products  $n_1 n_3$  and  $n_2 n_4$ . Hence both gcds are 1.

Conversely, suppose

$$\text{gcd}(n_1 n_2, n_3 n_4) = 1 \quad \text{and} \quad \text{gcd}(n_1 n_3, n_2 n_4) = 1.$$

The first gcd rules out common prime factors between any number in

$$\{n_1, n_2\}$$

and any number in

$$\{n_3, n_4\}.$$

So it rules out common factors for

$$(n_1, n_3), (n_1, n_4), (n_2, n_3), (n_2, n_4).$$

The second gcd rules out common factors for

$$(n_1, n_2), (n_1, n_4), (n_3, n_2), (n_3, n_4).$$

Together, every pair is covered. Thus every pair has gcd 1, so the four integers are pairwise relatively prime.

For  $k$  integers, assign each index a distinct binary label using

$$\lceil \lg k \rceil$$

bits. For each bit position, form two products: one product of all  $n_i$ 's whose label has bit 0, and one product of all  $n_i$ 's whose label has bit 1. Test whether the two products are relatively prime.

Any two distinct labels differ in some bit position, so every pair of numbers is separated by at least one such test. Therefore,

|                                                                                  |
|----------------------------------------------------------------------------------|
| $\lceil \lg k \rceil$ gcd tests are enough to check pairwise relative primality. |
|----------------------------------------------------------------------------------|

# Modular Groups Exercises

Q&A Notes

## Question 1

Draw the group operation tables for  $(\mathbb{Z}_4, +_4)$  and  $(\mathbb{Z}_5^*, \cdot_5)$ . Show that these groups are isomorphic by giving a one-to-one correspondence  $\alpha$  such that

$$a + b \equiv c \pmod{4}$$

if and only if

$$\alpha(a)\alpha(b) \equiv \alpha(c) \pmod{5}.$$

## Answer 1

The two groups are isomorphic. The key correspondence is

$$\alpha(0) = 1, \quad \alpha(1) = 2, \quad \alpha(2) = 4, \quad \alpha(3) = 3.$$

### Step 1: Build the table for $(\mathbb{Z}_4, +_4)$

Here

$$\mathbb{Z}_4 = \{0, 1, 2, 3\}.$$

The operation is: add normally, then take the remainder modulo 4.

For example,

$$2 + 3 = 5 \equiv 1 \pmod{4}.$$

So the entry in row 2, column 3, is 1.

The full operation table is:

| $+_4$ | 0 | 1 | 2 | 3 |
|-------|---|---|---|---|
| 0     | 0 | 1 | 2 | 3 |
| 1     | 1 | 2 | 3 | 0 |
| 2     | 2 | 3 | 0 | 1 |
| 3     | 3 | 0 | 1 | 2 |

### Step 2: Build the table for $(\mathbb{Z}_5^*, \cdot_5)$

By definition,

$$\mathbb{Z}_5^* = \{a : 1 \leq a < 5, \gcd(a, 5) = 1\}.$$

Since 5 is prime, every nonzero residue modulo 5 is relatively prime to 5. Therefore

$$\mathbb{Z}_5^* = \{1, 2, 3, 4\},$$

The operation is: multiply normally, then take the remainder modulo 5.

For example,

$$3 \cdot 4 = 12 \equiv 2 \pmod{5}.$$

So the entry in row 3, column 4, is 2.

The full operation table is:

| $\cdot_5$ | 1 | 2 | 3 | 4 |
|-----------|---|---|---|---|
| 1         | 1 | 2 | 3 | 4 |
| 2         | 2 | 4 | 1 | 3 |
| 3         | 3 | 1 | 4 | 2 |
| 4         | 4 | 3 | 2 | 1 |

### Step 3: Choose the matching

We need a one-to-one map

$$\alpha : \mathbb{Z}_4 \rightarrow \mathbb{Z}_5^*.$$

The useful matching comes from powers of 2 modulo 5:

$$2^0 \equiv 1, \quad 2^1 \equiv 2, \quad 2^2 \equiv 4, \quad 2^3 \equiv 3 \pmod{5}.$$

So

$$\boxed{\alpha(k) = 2^k \pmod{5}.}$$

This gives

$$0 \mapsto 1, \quad 1 \mapsto 2, \quad 2 \mapsto 4, \quad 3 \mapsto 3.$$

### Step 4: Check that addition becomes multiplication

We must show that

$$a + b \equiv c \pmod{4}$$

if and only if

$$\alpha(a)\alpha(b) \equiv \alpha(c) \pmod{5}.$$

Since

$$\alpha(a) = 2^a \quad \text{and} \quad \alpha(b) = 2^b,$$

Then

$$\alpha(a)\alpha(b) = 2^a 2^b = 2^{a+b}.$$

If

$$a + b \equiv c \pmod{4},$$

then  $a + b = c + 4t$  for some integer  $t$ . Hence

$$2^{a+b} = 2^{c+4t} = 2^c (2^4)^t.$$

But

$$2^4 = 16 \equiv 1 \pmod{5}.$$

Therefore

$$2^{a+b} \equiv 2^c \pmod{5}.$$

So

$$\alpha(a)\alpha(b) \equiv \alpha(c) \pmod{5}.$$

### Concrete check

Take

$$a = 2, \quad b = 3.$$

In  $\mathbb{Z}_4$ ,

$$2 + 3 = 5 \equiv 1 \pmod{4}.$$

So  $c = 1$ .

Apply  $\alpha$ :

$$\alpha(2) = 4, \quad \alpha(3) = 3, \quad \alpha(1) = 2.$$

Now check multiplication modulo 5:

$$\alpha(2)\alpha(3) = 4 \cdot 3 = 12 \equiv 2 \pmod{5}.$$

Since

$$\alpha(1) = 2,$$

we get

$$\alpha(2)\alpha(3) \equiv \alpha(1) \pmod{5}.$$

Therefore,

$$\boxed{(\mathbb{Z}_4, +_4) \cong (\mathbb{Z}_5^*, \cdot_5)}.$$

## Question 2

List all subgroups of  $\mathbb{Z}_9$  and of  $\mathbb{Z}_{13}^*$ .

## Answer 2

### Part A: Subgroups of $\mathbb{Z}_9$

The group is

$$\mathbb{Z}_9 = \{0, 1, 2, 3, 4, 5, 6, 7, 8\},$$

with addition modulo 9.

A subgroup of  $\mathbb{Z}_9$  must contain the identity 0, must be closed under addition modulo 9, and its size must divide the group size.

Since

$$|\mathbb{Z}_9| = 9,$$

the possible subgroup sizes are the divisors of 9:

$$1, 3, 9.$$

#### Size 1

The subgroup of size 1 contains only the identity:

$$\boxed{\{0\}}.$$

#### Size 3

Start with 3 and keep adding 3 modulo 9:

$$3,$$

$$3 + 3 = 6,$$

$$6 + 3 = 9 \equiv 0 \pmod{9}.$$

So

$$\boxed{\langle 3 \rangle = \{0, 3, 6\}}.$$

#### Size 9

Start with 1. It generates every element of  $\mathbb{Z}_9$ :

$$1, 2, 3, 4, 5, 6, 7, 8, 0.$$

So

$$\boxed{\langle 1 \rangle = \mathbb{Z}_9 = \{0, 1, 2, 3, 4, 5, 6, 7, 8\}}.$$

Therefore, all subgroups of  $\mathbb{Z}_9$  are

$$\{0\}, \quad \{0, 3, 6\}, \quad \{0, 1, 2, 3, 4, 5, 6, 7, 8\}.$$

## Part B: Subgroups of $\mathbb{Z}_{13}^*$

First calculate the group:

$$\mathbb{Z}_{13}^* = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}.$$

Since 13 is prime, every number from 1 to 12 is relatively prime to 13.

So

$$|\mathbb{Z}_{13}^*| = 12.$$

By Lagrange's theorem, subgroup sizes must divide 12:

$$1, 2, 3, 4, 6, 12.$$

Now use powers of 2 modulo 13:

$$\begin{aligned} 2^1 &\equiv 2, & 2^2 &\equiv 4, & 2^3 &\equiv 8, & 2^4 &\equiv 3, \\ 2^5 &\equiv 6, & 2^6 &\equiv 12, & 2^7 &\equiv 11, & 2^8 &\equiv 9, \\ 2^9 &\equiv 5, & 2^{10} &\equiv 10, & 2^{11} &\equiv 7, & 2^{12} &\equiv 1 \pmod{13}. \end{aligned}$$

Thus

$$\langle 2 \rangle = \mathbb{Z}_{13}^*.$$

So  $\mathbb{Z}_{13}^*$  is cyclic, and its subgroups come from powers of the generator 2.

### Size 1

$$\{1\}.$$

### Size 2

Use  $2^6 \equiv 12 \pmod{13}$ :

$$12^1 \equiv 12, \quad 12^2 = 144 \equiv 1 \pmod{13}.$$

So

$$\{1, 12\}.$$

### Size 3

Use  $2^4 \equiv 3 \pmod{13}$ :

$$3^1 \equiv 3, \quad 3^2 = 9, \quad 3^3 = 27 \equiv 1 \pmod{13}.$$

So

$$\{1, 3, 9\}.$$

### Size 4

Use  $2^3 \equiv 8 \pmod{13}$ :

$$\begin{aligned} 8^1 &\equiv 8, & 8^2 &= 64 \equiv 12, \\ 8^3 &= 512 \equiv 5, & 8^4 &= 4096 \equiv 1 \pmod{13}. \end{aligned}$$

So

$$\{1, 5, 8, 12\}.$$

### Size 6

Use  $2^2 \equiv 4 \pmod{13}$ :

$$\begin{aligned} 4^1 &\equiv 4, & 4^2 &= 16 \equiv 3, & 4^3 &= 64 \equiv 12, \\ 4^4 &= 256 \equiv 9, & 4^5 &= 1024 \equiv 10, & 4^6 &= 4096 \equiv 1 \pmod{13}. \end{aligned}$$

So

$$\{1, 3, 4, 9, 10, 12\}.$$

### Size 12

The whole group is the subgroup of size 12:

$$\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}.$$

#### Main idea

subgroup sizes must divide the group size.

For  $\mathbb{Z}_9$ , the group size is 9, so the possible sizes are 1, 3, 9. For  $\mathbb{Z}_{13}^*$ , the group size is 12, so the possible sizes are 1, 2, 3, 4, 6, 12.

### Question 3

Prove the subgroup test: if  $(S, \oplus)$  is a finite group and  $S'$  is a nonempty subset of  $S$  such that

$$a \oplus b \in S'$$

for all  $a, b \in S'$ , then  $(S', \oplus)$  is a subgroup of  $(S, \oplus)$ .

### Answer 3

We need to show that  $S'$  has the group properties using the same operation  $\oplus$ .

Closure is already given:

$$a, b \in S' \implies a \oplus b \in S'.$$

Associativity is inherited from the bigger group  $S$ .

Now take any element  $a \in S'$ . Since  $S'$  is closed,

$$a, \quad a^2, \quad a^3, \quad \dots$$

all belong to  $S'$ , where  $a^k$  means applying the operation  $k$  times.

Because  $S'$  is finite, two powers must repeat:

$$a^i = a^j$$

for some  $i < j$ . Cancel  $a^i$  inside the bigger group  $S$ :

$$e = a^{j-i}.$$

So the identity element  $e$  belongs to  $S'$ .

Also,

$$a^{j-i} = e.$$

Thus

$$a \oplus a^{j-i-1} = e,$$

so  $a^{j-i-1}$  is the inverse of  $a$ . Since  $S'$  is closed, this inverse also belongs to  $S'$ .

Therefore  $S'$  contains the identity, contains inverses, is closed, and is associative. Hence

$$\boxed{S' \text{ is a subgroup of } S.}$$

#### Question 4

Show that if  $p$  is prime and  $e$  is a positive integer, then

$$\phi(p^e) = p^{e-1}(p-1).$$

#### Answer 4

The numbers not relatively prime to  $p^e$  are exactly the multiples of  $p$ .

There are  $p^e$  residues modulo  $p^e$ :

$$0, 1, 2, \dots, p^e - 1.$$

The multiples of  $p$  among them are

$$0, p, 2p, \dots, (p^{e-1} - 1)p.$$

So there are

$$p^{e-1}$$

multiples of  $p$ .

Therefore the number of residues relatively prime to  $p^e$  is

$$p^e - p^{e-1}.$$

Factor:

$$p^e - p^{e-1} = p^{e-1}(p-1).$$

Hence

$$\boxed{\phi(p^e) = p^{e-1}(p-1).}$$

#### Question 5

Show that for any integer  $n > 1$  and for any  $a \in \mathbb{Z}_n^*$ , the function

$$f_a : \mathbb{Z}_n^* \rightarrow \mathbb{Z}_n^*$$

defined by

$$f_a(x) = ax \bmod n$$

is a permutation of  $\mathbb{Z}_n^*$ .

#### Answer 5

The function is a permutation because multiplication by an invertible element can be undone. Since

$$a \in \mathbb{Z}_n^*,$$

we have

$$\gcd(a, n) = 1.$$

So  $a$  has an inverse modulo  $n$ , call it  $a^{-1}$ .

First,  $f_a$  maps  $\mathbb{Z}_n^*$  back into  $\mathbb{Z}_n^*$ . If  $x \in \mathbb{Z}_n^*$ , then

$$\gcd(x, n) = 1.$$

Since both  $a$  and  $x$  are relatively prime to  $n$ ,

$$\gcd(ax, n) = 1.$$

So

$$ax \bmod n \in \mathbb{Z}_n^*.$$

Now show the function is one-to-one. Suppose

$$f_a(x) = f_a(y).$$

Then

$$ax \equiv ay \pmod{n}.$$

Multiply both sides by  $a^{-1}$ :

$$x \equiv y \pmod{n}.$$

So different inputs cannot give the same output.

Since  $\mathbb{Z}_n^*$  is finite, a one-to-one function from  $\mathbb{Z}_n^*$  to itself is also onto. Therefore  $f_a$  rearranges the elements of  $\mathbb{Z}_n^*$ , meaning it is a permutation:

|                                              |
|----------------------------------------------|
| $f_a$ is a permutation of $\mathbb{Z}_n^*$ . |
|----------------------------------------------|

# Modular Linear Equations Exercises

Q&A Notes

## Question 1

Find all solutions to the equation

$$35x \equiv 10 \pmod{50}.$$

## Answer 1

The solutions are

$$x \equiv 6, 16, 26, 36, 46 \pmod{50}.$$

Let

$$a = 35, \quad b = 10, \quad n = 50.$$

First compute

$$d = \gcd(35, 50) = 5.$$

Since

$$5 \mid 10,$$

solutions exist. There will be exactly

$$\boxed{5}$$

solutions modulo 50.

Now use Extended Euclid:

$$50 = 35(1) + 15,$$

$$35 = 15(2) + 5.$$

Back-substitute:

$$5 = 35 - 15(2).$$

Since

$$15 = 50 - 35,$$

we get

$$5 = 35 - 2(50 - 35) = 3(35) - 2(50).$$

Thus

$$5 = 35(3) + 50(-2),$$

so

$$x' = 3.$$

One solution is

$$x_0 = x' \left( \frac{b}{d} \right) \pmod{n}.$$

Therefore

$$x_0 = 3 \left( \frac{10}{5} \right) \pmod{50} = 6.$$

The spacing between solutions is

$$\frac{n}{d} = \frac{50}{5} = 10.$$

So the 5 solutions are

$$x_i = 6 + 10i \pmod{50}, \quad i = 0, 1, 2, 3, 4.$$

Hence

$$x \equiv 6, 16, 26, 36, 46 \pmod{50}.$$

### Question 2

Prove that

$$ax \equiv ay \pmod{n}$$

implies

$$x \equiv y \pmod{n}$$

whenever  $\gcd(a, n) = 1$ . Show that the condition  $\gcd(a, n) = 1$  is necessary by supplying a counterexample with  $\gcd(a, n) > 1$ .

### Answer 2

If  $\gcd(a, n) = 1$ , then cancellation is valid modulo  $n$ .

From

$$ax \equiv ay \pmod{n},$$

we get

$$n \mid (ax - ay).$$

Factor:

$$n \mid a(x - y).$$

Since

$$\gcd(a, n) = 1,$$

Euclid's lemma gives

$$n \mid (x - y).$$

Therefore

$$\boxed{x \equiv y \pmod{n}}.$$

Now show why the condition is necessary. Take

$$a = 2, \quad n = 6, \quad x = 1, \quad y = 4.$$

Then

$$\gcd(2, 6) = 2 > 1.$$

Also,

$$ax = 2(1) = 2,$$

and

$$ay = 2(4) = 8 \equiv 2 \pmod{6}.$$

So

$$ax \equiv ay \pmod{6}.$$

But

$$1 \not\equiv 4 \pmod{6}.$$

Thus cancellation fails when  $\gcd(a, n) > 1$ .

### Takeaway

$\boxed{\text{You may cancel } a \text{ modulo } n \text{ only when } \gcd(a, n) = 1.}$

### Question 3

In the procedure `ModularLinearEquationSolver`, consider changing line 3 to

$$x_0 = x' \left( \frac{b}{d} \right) \bmod \left( \frac{n}{d} \right).$$

Will this work? Explain why or why not.

### Answer 3

Yes, this change works. It prints the same solution set, possibly in a different order.

The original line computes

$$x_0 = x' \left( \frac{b}{d} \right) \bmod n.$$

The proposed line computes the same value modulo  $n/d$ :

$$x_0 = x' \left( \frac{b}{d} \right) \bmod \left( \frac{n}{d} \right).$$

This is valid because all solutions are spaced by

$$\frac{n}{d}.$$

If one solution is changed by a multiple of  $n/d$ , it is still one of the same solution family.

More directly, suppose  $x_0$  is a solution. Let

$$x'_0 = x_0 - q \left( \frac{n}{d} \right)$$

for some integer  $q$ . Then

$$ax'_0 = ax_0 - q \frac{an}{d}.$$

Since  $d \mid a$ , the number  $a/d$  is an integer. Hence

$$q \frac{an}{d} = q \left( \frac{a}{d} \right) n$$

is a multiple of  $n$ . Therefore

$$ax'_0 \equiv ax_0 \pmod{n}.$$

So  $x'_0$  is also a solution.

### Example

For

$$14x \equiv 30 \pmod{100},$$

we have

$$d = 2, \quad x' = -7.$$

The old line gives

$$x_0 = (-7)(15) \bmod 100 = 95.$$

The changed line gives

$$x_0 = (-7)(15) \bmod 50 = 45.$$

The solution spacing is

$$\frac{100}{2} = 50.$$

So the old line prints

$$95, 45,$$

while the changed line prints

$$45, 95.$$

The same solutions appear.

Thus

the modified line works.

### Question 4

Let  $p$  be prime and

$$f(x) \equiv f_0 + f_1x + \cdots + f_tx^t \pmod{p}$$

be a polynomial of degree  $t$ , with coefficients drawn from  $\mathbb{Z}_p$ . A value  $a \in \mathbb{Z}_p$  is a zero of  $f$  if

$$f(a) \equiv 0 \pmod{p}.$$

Prove that if  $a$  is a zero of  $f$ , then

$$f(x) \equiv (x - a)g(x) \pmod{p}$$

for some polynomial  $g(x)$  of degree  $t - 1$ . Prove by induction on  $t$  that a nonzero polynomial of degree  $t$  has at most  $t$  distinct zeros modulo  $p$ .

### Answer 4

Because  $p$  is prime,  $\mathbb{Z}_p$  behaves like a field: every nonzero element has a multiplicative inverse.

#### Part 1: Factor theorem modulo $p$

Divide  $f(x)$  by  $x - a$  in  $\mathbb{Z}_p$ . Polynomial division gives

$$f(x) \equiv (x - a)g(x) + r \pmod{p},$$

where  $r$  is a constant remainder.

Now substitute  $x = a$ :

$$f(a) \equiv (a - a)g(a) + r \pmod{p}.$$

So

$$f(a) \equiv r \pmod{p}.$$

Since  $a$  is a zero,

$$f(a) \equiv 0 \pmod{p}.$$

Therefore

$$r \equiv 0 \pmod{p}.$$

Hence

$$f(x) \equiv (x - a)g(x) \pmod{p}.$$

Since  $f$  has degree  $t$ , the polynomial  $g$  has degree  $t - 1$ .

## Part 2: Root bound by induction

We prove: a nonzero polynomial of degree  $t$  has at most  $t$  distinct zeros modulo  $p$ .

**Base case:** If  $t = 0$ , then  $f(x)$  is a nonzero constant. A nonzero constant is never 0 modulo  $p$ , so it has 0 roots. This is at most 0.

**Inductive step:** Assume every nonzero polynomial of degree  $t - 1$  has at most  $t - 1$  distinct zeros modulo  $p$ .

Let  $f(x)$  have degree  $t$ . If  $f$  has no zeros, then the result is done. Otherwise, let  $a$  be one zero of  $f$ . By Part 1,

$$f(x) \equiv (x - a)g(x) \pmod{p},$$

where  $g(x)$  has degree  $t - 1$ .

Now let  $b \neq a$  be another zero of  $f$ . Then

$$0 \equiv f(b) \equiv (b - a)g(b) \pmod{p}.$$

Because  $b \neq a$ ,

$$b - a \not\equiv 0 \pmod{p}.$$

Since  $p$  is prime,  $b - a$  has an inverse modulo  $p$ . Therefore we can cancel  $b - a$ , giving

$$g(b) \equiv 0 \pmod{p}.$$

So every zero of  $f$  except  $a$  is a zero of  $g$ .

By the induction hypothesis,  $g$  has at most  $t - 1$  zeros. Including  $a$ , the polynomial  $f$  has at most

$$1 + (t - 1) = t$$

zeros.

Therefore,

$$\boxed{\text{a nonzero polynomial of degree } t \text{ has at most } t \text{ distinct zeros modulo } p.}$$

# Chinese Remainder Theorem Exercises

Q&A Notes

## Question 1

Find all solutions to the equations

$$x \equiv 4 \pmod{5}$$

and

$$x \equiv 5 \pmod{11}.$$

## Answer 1

The solution is

$$x \equiv 49 \pmod{55}.$$

So all integer solutions are

$$x = 49 + 55t, \quad t \in \mathbb{Z}.$$

The moduli 5 and 11 are relatively prime, so the Chinese Remainder Theorem applies. First compute the combined modulus:

$$N = 5 \cdot 11 = 55.$$

For the congruence modulo 5:

$$M_1 = \frac{55}{5} = 11.$$

We need  $y_1$  such that

$$11y_1 \equiv 1 \pmod{5}.$$

Since

$$11 \equiv 1 \pmod{5},$$

we can take

$$y_1 = 1.$$

For the congruence modulo 11:

$$M_2 = \frac{55}{11} = 5.$$

We need  $y_2$  such that

$$5y_2 \equiv 1 \pmod{11}.$$

Since

$$5 \cdot 9 = 45 \equiv 1 \pmod{11},$$

we can take

$$y_2 = 9.$$

Now combine:

$$x \equiv 4M_1y_1 + 5M_2y_2 \pmod{55}.$$

$$x \equiv 4(11)(1) + 5(5)(9) \pmod{55}.$$

$$x \equiv 44 + 225 \pmod{55}.$$

$$x \equiv 269 \pmod{55}.$$

$$269 = 55(4) + 49.$$

Therefore

$$x \equiv 49 \pmod{55}.$$

Check:

$$49 \equiv 4 \pmod{5}, \quad 49 \equiv 5 \pmod{11}.$$

## Question 2

Find all integers  $x$  that leave remainders 1, 2, 3 when divided by 9, 8, 7, respectively.

## Answer 2

The solution is

$$x \equiv 10 \pmod{504}.$$

So all integer solutions are

$$x = 10 + 504t, \quad t \in \mathbb{Z}.$$

The system is

$$x \equiv 1 \pmod{9}, \quad x \equiv 2 \pmod{8}, \quad x \equiv 3 \pmod{7}.$$

The moduli 9, 8, 7 are pairwise relatively prime, so CRT applies.

Compute

$$N = 9 \cdot 8 \cdot 7 = 504.$$

| Congruence | $n_i$ | $a_i$ | $M_i = N/n_i$ |
|------------|-------|-------|---------------|
| 1          | 9     | 1     | 56            |
| 2          | 8     | 2     | 63            |
| 3          | 7     | 3     | 72            |

Now find the inverses.

For  $M_1 = 56$ , work modulo 9:

$$56 \equiv 2 \pmod{9}.$$

We need

$$2y_1 \equiv 1 \pmod{9}.$$

Since

$$2 \cdot 5 = 10 \equiv 1 \pmod{9},$$

we take

$$y_1 = 5.$$

For  $M_2 = 63$ , work modulo 8:

$$63 \equiv 7 \pmod{8}.$$

We need

$$7y_2 \equiv 1 \pmod{8}.$$

Since

$$7 \cdot 7 = 49 \equiv 1 \pmod{8},$$

we take

$$y_2 = 7.$$

For  $M_3 = 72$ , work modulo 7:

$$72 \equiv 2 \pmod{7}.$$

We need

$$2y_3 \equiv 1 \pmod{7}.$$

Since

$$2 \cdot 4 = 8 \equiv 1 \pmod{7},$$

we take

$$y_3 = 4.$$

Combine:

$$x \equiv a_1 M_1 y_1 + a_2 M_2 y_2 + a_3 M_3 y_3 \pmod{504}.$$

$$x \equiv 1(56)(5) + 2(63)(7) + 3(72)(4) \pmod{504}.$$

$$x \equiv 280 + 882 + 864 \pmod{504}.$$

$$x \equiv 2026 \pmod{504}.$$

Reduce:

$$2026 = 504(4) + 10.$$

Therefore

$$\boxed{x \equiv 10 \pmod{504}.$$

Check:

$$10 \equiv 1 \pmod{9}, \quad 10 \equiv 2 \pmod{8}, \quad 10 \equiv 3 \pmod{7}.$$

### Question 3

Argue that, under the definitions of the Chinese Remainder Theorem, if  $\gcd(a, n) = 1$ , then the inverse of  $a$  modulo  $n$  corresponds to the inverses of its parts modulo  $n_1, n_2, \dots, n_k$ .

### Answer 3

The inverse modulo  $n$  reduces to the inverse in each smaller modulus.

Assume

$$n = n_1 n_2 \cdots n_k,$$

where the  $n_i$ 's are pairwise relatively prime.

Let

$$b \equiv a^{-1} \pmod{n}.$$

This means

$$ab \equiv 1 \pmod{n}.$$

If a number is congruent to 1 modulo  $n$ , then it is also congruent to 1 modulo every divisor  $n_i$  of  $n$ . Hence

$$ab \equiv 1 \pmod{n_i}$$

for every  $i$ .

Now reduce  $a$  and  $b$  modulo  $n_i$ . Write

$$a_i \equiv a \pmod{n_i}, \quad b_i \equiv b \pmod{n_i}.$$

Then

$$a_i b_i \equiv 1 \pmod{n_i}.$$

So  $b_i$  is the inverse of  $a_i$  modulo  $n_i$ .

Therefore

$$\boxed{a^{-1} \pmod{n} \text{ corresponds to } (a_1^{-1} \pmod{n_1}, \dots, a_k^{-1} \pmod{n_k}).}$$

### Why this makes sense

CRT lets us work modulo  $n$  by working separately modulo  $n_1, n_2, \dots, n_k$ . Multiplicative inverses also split this way.

### Question 4

Under the definitions of the Chinese Remainder Theorem, prove that for any polynomial  $f$ , the number of roots of

$$f(x) \equiv 0 \pmod{n}$$

equals the product of the numbers of roots of

$$f(x) \equiv 0 \pmod{n_1}, \quad f(x) \equiv 0 \pmod{n_2}, \quad \dots, \quad f(x) \equiv 0 \pmod{n_k},$$

where  $n = n_1 n_2 \cdots n_k$  and the  $n_i$ 's are pairwise relatively prime.

### Answer 4

The number of roots modulo  $n$  is the product of the numbers of roots modulo the smaller moduli.

Let

$$R_i = \{r \pmod{n_i} : f(r) \equiv 0 \pmod{n_i}\}$$

be the set of roots modulo  $n_i$ .

Suppose

$$x \equiv r_i \pmod{n_i}$$

for each  $i$ . By CRT, every choice of residues

$$(r_1, r_2, \dots, r_k)$$

comes from exactly one residue  $x$  modulo  $n$ .

Now,

$$f(x) \equiv 0 \pmod{n}$$

is equivalent to saying

$$n \mid f(x).$$

Since

$$n = n_1 n_2 \cdots n_k$$

and the  $n_i$ 's are pairwise relatively prime, this is equivalent to

$$n_i \mid f(x)$$

for every  $i$ . In congruence form:

$$f(x) \equiv 0 \pmod{n_i} \quad \text{for every } i.$$

So  $x$  is a root modulo  $n$  exactly when its residue modulo each  $n_i$  is a root modulo  $n_i$ .

Therefore the roots modulo  $n$  are obtained by independently choosing:

$$r_1 \in R_1, \quad r_2 \in R_2, \quad \dots, \quad r_k \in R_k.$$

If

$$|R_1| = s_1, \quad |R_2| = s_2, \quad \dots, \quad |R_k| = s_k,$$

then the number of possible choices is

$$s_1 s_2 \cdots s_k.$$

Hence

$$\#\{x \pmod{n} : f(x) \equiv 0 \pmod{n}\} = \prod_{i=1}^k \#\{x \pmod{n_i} : f(x) \equiv 0 \pmod{n_i}\}.$$

# Powers and Modular Exponentiation

Q&A Notes

## Question 1

Draw a table showing the order of every element in  $\mathbb{Z}_{11}^*$ . Pick the smallest primitive root  $g$  and compute a table giving  $\text{ind}_{11,g}(x)$  for all  $x \in \mathbb{Z}_{11}^*$ .

## Answer 1

The smallest primitive root modulo 11 is

$$g = 2.$$

The group  $\mathbb{Z}_{11}^*$  contains the nonzero residues modulo 11:

$$\mathbb{Z}_{11}^* = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}.$$

The order of an element  $a$  is the smallest positive integer  $t$  such that

$$a^t \equiv 1 \pmod{11}.$$

| Element | Order | Reason                                    |
|---------|-------|-------------------------------------------|
| 1       | 1     | $1^1 \equiv 1 \pmod{11}$ .                |
| 2       | 10    | Powers are 2, 4, 8, 5, 10, 9, 7, 3, 6, 1. |
| 3       | 5     | Powers are 3, 9, 5, 4, 1.                 |
| 4       | 5     | Powers are 4, 5, 9, 3, 1.                 |
| 5       | 5     | Powers are 5, 3, 4, 9, 1.                 |
| 6       | 10    | Powers are 6, 3, 7, 9, 10, 5, 8, 4, 2, 1. |
| 7       | 10    | Powers are 7, 5, 2, 3, 10, 4, 6, 9, 8, 1. |
| 8       | 10    | Powers are 8, 9, 6, 4, 10, 3, 2, 5, 7, 1. |
| 9       | 5     | Powers are 9, 4, 3, 5, 1.                 |
| 10      | 2     | $10^2 = 100 \equiv 1 \pmod{11}$ .         |

A primitive root modulo 11 is an element with order

$$\phi(11) = 10.$$

The smallest element with order 10 is 2, so

$$g = 2.$$

Now compute the index table. The value

$$\text{ind}_{11,2}(x)$$

means the exponent  $k$  such that

$$2^k \equiv x \pmod{11}.$$

| $k$             | 0 | 1 | 2 | 3 | 4 | 5  | 6 | 7 | 8 | 9 |
|-----------------|---|---|---|---|---|----|---|---|---|---|
| $2^k \pmod{11}$ | 1 | 2 | 4 | 8 | 5 | 10 | 9 | 7 | 3 | 6 |

So the index table is:

| $x$ | $\text{ind}_{11,2}(x)$ |
|-----|------------------------|
| 1   | 0                      |
| 2   | 1                      |
| 3   | 8                      |
| 4   | 2                      |
| 5   | 4                      |
| 6   | 9                      |
| 7   | 7                      |
| 8   | 3                      |
| 9   | 6                      |
| 10  | 5                      |

### Answer Summary

$$g = 2$$

and

$$\text{ind}_{11,2}(1, 2, 3, 4, 5, 6, 7, 8, 9, 10) = (0, 1, 8, 2, 4, 9, 7, 3, 6, 5).$$

### Question 2

Give a modular exponentiation algorithm that examines the bits of  $b$  from right to left instead of left to right.

### Answer 2

A right-to-left modular exponentiation algorithm keeps two values:

| Variable | Meaning                                               |
|----------|-------------------------------------------------------|
| $d$      | Accumulates the answer.                               |
| $base$   | Stores the current power of $a$ , repeatedly squared. |

### Right-to-Left Modular Exponentiation

```

RIGHT-TO-LEFT-MODULAR-EXPONENTIATION(a, b, n):
    d = 1
    base = a mod n

    while b > 0:
        if b is odd:
            d = (d * base) mod n

        base = (base * base) mod n
        b = floor(b / 2)

    return d

```

Why this works:

The binary expansion of  $b$  is

$$b = b_0 + 2b_1 + 4b_2 + \cdots + 2^k b_k,$$

where each bit  $b_i$  is 0 or 1.

So

$$a^b = a^{b_0} \cdot a^{2b_1} \cdot a^{4b_2} \cdots a^{2^k b_k}.$$

The algorithm scans the bits from right to left:

- If the current bit is 1, multiply the answer by the current base.
- Square the base each time, so it becomes  $a, a^2, a^4, a^8, \dots$
- Divide  $b$  by 2 to move to the next bit.

### Key Difference

Left-to-right exponentiation reads the binary representation from the most significant bit.

Right-to-left exponentiation reads it from the least significant bit using repeated division by 2.

### Question 3

Assuming that you know  $\phi(n)$ , explain how to compute

$$a^{-1} \bmod n$$

for any  $a \in \mathbb{Z}_n^*$  using the procedure MODULAR-EXPONENTIATION.

### Answer 3

The inverse is

$$a^{-1} \equiv a^{\phi(n)-1} \pmod{n}.$$

Since

$$a \in \mathbb{Z}_n^*,$$

we know

$$\gcd(a, n) = 1.$$

By Euler's theorem,

$$a^{\phi(n)} \equiv 1 \pmod{n}.$$

Rewrite the left side:

$$a^{\phi(n)} = a \cdot a^{\phi(n)-1}.$$

So

$$a \cdot a^{\phi(n)-1} \equiv 1 \pmod{n}.$$

This means  $a^{\phi(n)-1}$  is the multiplicative inverse of  $a$  modulo  $n$ :

$$a^{-1} \equiv a^{\phi(n)-1} \pmod{n}.$$

Therefore, to compute the inverse, run:

## Inverse by Modular Exponentiation

```
MODULAR-INVERSE-BY-EULER(a, n):  
    return MODULAR-EXPONENTIATION(a, phi(n) - 1, n)
```

### Example

Find

$$5^{-1} \bmod 12.$$

First compute

$$\phi(12) = 4.$$

Then

$$5^{-1} \equiv 5^{4-1} \equiv 5^3 \pmod{12}.$$

Compute:

$$5^2 = 25 \equiv 1 \pmod{12}.$$

So

$$5^3 \equiv 5 \pmod{12}.$$

Therefore

$$\boxed{5^{-1} \equiv 5 \pmod{12}.$$

Check:

$$5 \cdot 5 = 25 \equiv 1 \pmod{12}.$$

# RSA Cryptography Exercises

Q&A Notes

## Question 1

Consider an RSA key set with

$$p = 11, \quad q = 29, \quad n = 319, \quad e = 3.$$

What value of  $d$  should be used in the secret key? What is the encryption of the message  $M = 100$ ?

## Answer 1

The secret exponent is

$$d = 187$$

and the encryption of  $M = 100$  is

$$254.$$

First compute Euler's totient of  $n = pq$ :

$$\phi(n) = (p - 1)(q - 1) = (11 - 1)(29 - 1) = 10 \cdot 28 = 280.$$

The secret exponent  $d$  must satisfy

$$ed \equiv 1 \pmod{\phi(n)}.$$

Since  $e = 3$ , we need

$$3d \equiv 1 \pmod{280}.$$

Using Extended Euclid,

$$280 = 3(93) + 1,$$

so

$$1 = 280 - 3(93).$$

Thus

$$-93$$

is an inverse of 3 modulo 280. Reduce it:

$$-93 \equiv 187 \pmod{280}.$$

Therefore

$$d = 187.$$

Now encrypt  $M = 100$  using

$$C \equiv M^e \pmod{n}.$$

So

$$C \equiv 100^3 \pmod{319}.$$

Compute step by step:

$$100^2 = 10000 \equiv 111 \pmod{319},$$

because

$$10000 = 319(31) + 111.$$

Then

$$100^3 \equiv 111 \cdot 100 = 11100 \pmod{319}.$$

Since

$$11100 = 319(34) + 254,$$

we get

$$100^3 \equiv 254 \pmod{319}.$$

### Final RSA Values

Public key:  $(e, n) = (3, 319)$ .

Secret key:  $(d, n) = (187, 319)$ .

Ciphertext for  $M = 100$ :  $C = 254$ .

### Question 2

Prove that if Alice's public exponent is  $e = 3$  and an adversary obtains Alice's secret exponent  $d$ , where  $0 < d < \phi(n)$ , then the adversary can factor Alice's modulus  $n$  in polynomial time in the number of bits in  $n$ .

### Answer 2

If the adversary knows  $d$ , then they can recover  $\phi(n)$ , and once  $\phi(n)$  is known, factoring  $n = pq$  is easy.

In RSA,  $d$  is chosen so that

$$ed \equiv 1 \pmod{\phi(n)}.$$

Here  $e = 3$ , so

$$3d \equiv 1 \pmod{\phi(n)}.$$

That means

$$3d - 1 = k\phi(n)$$

for some positive integer  $k$ .

Because

$$0 < d < \phi(n),$$

we have

$$0 < 3d - 1 < 3\phi(n).$$

Therefore

$$k \in \{1, 2\}.$$

So the adversary tries the two possible values

$$\phi(n) = 3d - 1$$

or

$$\phi(n) = \frac{3d - 1}{2},$$

keeping the one that is an integer and gives valid factors.

Now suppose  $n = pq$ . Then

$$\phi(n) = (p - 1)(q - 1) = pq - p - q + 1.$$

Since  $pq = n$ ,

$$\phi(n) = n - (p + q) + 1.$$

Thus

$$p + q = n - \phi(n) + 1.$$

Let

$$s = p + q.$$

Then  $p$  and  $q$  are the two roots of

$$t^2 - st + n = 0.$$

So

$$t = \frac{s \pm \sqrt{s^2 - 4n}}{2}.$$

The adversary can compute  $s$ , compute the discriminant

$$\Delta = s^2 - 4n,$$

take its square root, and recover

$$p = \frac{s + \sqrt{\Delta}}{2}, \quad q = \frac{s - \sqrt{\Delta}}{2}.$$

### Why This Is Polynomial Time

Only a constant number of candidates for  $\phi(n)$  are tested. Arithmetic operations, integer square root, and checking the factors all take polynomial time in the number of bits of  $n$ .

### Question 3

Prove that RSA is multiplicative in the sense that

$$P_A(M_1)P_A(M_2) \equiv P_A(M_1M_2) \pmod{n}.$$

Use this fact to prove that if an adversary had a procedure that could efficiently decrypt 1% of messages from  $\mathbb{Z}_n$  encrypted with  $P_A$ , then the adversary could employ a probabilistic algorithm to decrypt every message encrypted with  $P_A$  with high probability.

### Answer 3

RSA is multiplicative because encryption is exponentiation modulo  $n$ .

Let Alice's public encryption function be

$$P_A(M) \equiv M^e \pmod{n}.$$

Then

$$P_A(M_1)P_A(M_2) \equiv M_1^e M_2^e \pmod{n}.$$

Using the exponent rule,

$$M_1^e M_2^e = (M_1 M_2)^e.$$

Therefore

$$P_A(M_1)P_A(M_2) \equiv (M_1 M_2)^e \equiv P_A(M_1 M_2) \pmod{n}.$$

So

$$P_A(M_1)P_A(M_2) \equiv P_A(M_1 M_2) \pmod{n}.$$

Now suppose an adversary has a procedure  $D$  that decrypts 1% of RSA ciphertexts efficiently. The adversary wants to decrypt a target ciphertext

$$C = P_A(M).$$

The adversary chooses a random invertible value

$$r \in \mathbb{Z}_n^*$$

and computes

$$C' = C \cdot P_A(r) \pmod{n}.$$

Using the multiplicative property,

$$C' \equiv P_A(M)P_A(r) \equiv P_A(Mr) \pmod{n}.$$

So  $C'$  is an encryption of the randomized message  $Mr$ .

If the decryption procedure succeeds on  $C'$ , it returns

$$Mr \pmod{n}.$$

Since  $r \in \mathbb{Z}_n^*$ , the inverse  $r^{-1} \pmod{n}$  exists. The adversary then computes

$$M \equiv (Mr)r^{-1} \pmod{n}.$$

Each random choice of  $r$  gives the decryption procedure a fresh-looking encrypted message. Since the procedure works on 1% of messages, one trial succeeds with probability 0.01. After  $T$  independent trials, the probability of failure is at most

$$(0.99)^T.$$

Thus, by repeating enough times, the adversary decrypts the original message with high probability.

### Main Idea

The adversary hides the target message  $M$  by multiplying it by a random invertible  $r$ . If the weak decryptor can decrypt  $P_A(Mr)$ , then multiplying by  $r^{-1}$  recovers  $M$ .